

Kyle Tran

Mechanical Engineering, Robotics minor | Northeastern '29 | Seeking a Summer 2027 hardware or robotics co-op
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EDUCATION

Northeastern University | *Boston, MA* Expected May 2029
Bachelor of Science in Mechanical Engineering, Minor in Robotics | GPA: 3.87/4.00
Relevant Coursework: Cornerstone of Engineering I, Statics, Thermodynamics, Materials Science, Probability and Statistics

PROJECTS

D.R.E.A.M.S Autonomous Environmental Survey Boat | *Personal Project* Jan 2026 – Present

- Designed a twin-hull catamaran in SolidWorks, concept to build: 3D-printed ASA hull, fiberglass-and-epoxy skin
- v1 completed autonomous waypoint surveys on local lakes: three RP2040 boards on a 433 MHz HC-12 radio link, and a winch that lowers a sensor pod to sample the water
- Presented v1 at the inaugural Creators' Showcase at Northeastern University London
- Designing the v2 KiCad carrier board (second revision, in bring-up) that replaces v1's hand-wired harness with three isolated battery domains, two VESC motor controllers and every sensor header
- Specified the v2 architecture (Raspberry Pi 5 plus two RP2040s, stereo vision, 4G telemetry), directed AI coding agents to implement the software, and lead integration and testing

Combat Robotics | *Team Dihydrogen Monoxide* Mar 2023 – Jun 2025

- Designed and built about ten robots across five weight classes in Onshape: spinner and drum weapons, brushless drive, aluminum frames sent out for CNC machining, 3D-printed nylon and polycarbonate armor
- 4th nationally in Plastic Antweight (2024 RCL National Championship); U.S. rankings of top-15 (1 lb), top-30 (1 lb plastic) and top-35 (3 lb); 4 first- and 8 second-place regional finishes
- Designed a single-cam shuffler drive so the newest version of Achilles (1 lb) walks instead of rolling

Autonomous Maze-Navigating Robot | *Cornerstone of Engineering I, with Matthew Hong* Oct 2025 – Dec 2025

- Designed a 3D-printed tank-drive chassis in Onshape to replace the class kit, which tipped in fast turns: 1.6 in tall, about half the kit's height, and every printed part fit the first time
- Wrote the MicroPython firmware: proportional control that centers the robot between walls using two angled ultrasonic sensors, an LCD program menu, and asyncio so animations never stall the motors
- Added crowned wheels to keep the rubber-band treads on in hard turns; completed all three challenge courses

Adaptive Compliant Robotic Arm | *Personal Project, In Design* 2026 – Present

- Five-axis arm; three cycloidal gearboxes (14:1 to 22:1) with output-side encoders, designed to hold $\pm 0.25^\circ$
- Built a gearbox simulator with AI coding agents; all three gearboxes pass its stress and bearing-life checks

LEADERSHIP EXPERIENCE

President | *Combat Robots Club, Burlingame, CA* Oct 2023 – Jun 2025

- Taught CAD, soldering, electronics and 3D printing; secured sponsorships and ran club finances and event travel
- Led the club to 16 podium finishes (4 first, 7 second, 5 third) across 5 events

Subsystem Lead | *Iron Panthers, FRC Team 5026, Burlingame, CA* Sep 2021 – May 2024

- Designed and prototyped the spindexer mechanism in CAD; led 10+ students, advised by 5 professional mentors

Ceramic Tools R&D | *Burlingame High School* Oct 2023 – May 2025

- Designed, 3D-printed and published 10+ tools for the ceramics classes, free for anyone to print on MakerWorld

SKILLS

CAD and design: SolidWorks, Onshape, Fusion 360; mechanism design, tolerance analysis, DFM, rapid prototyping

Fabrication: 2.5D CNC machining, CNC router, PCB mill, waterjet, laser cutter, FDM 3D printing, composite layup (fiberglass, carbon fiber), soldering

Electronics: embedded systems, RP2040 microcontrollers, Raspberry Pi, KiCad PCB design, motor control, sensors

Programming: Python, MicroPython (asyncio), MATLAB, Java, GitHub; AI coding agents (Claude Code)

AWARDS

4th, Robot Combat League National Championship, Plastic Antweight (2024) | 1st, CROS March Madness (2024) | 1st, BCRC Bell Brawls IV (2024) | 1st, Spartan Bell Slugfest (2024) | 1st, NorCal Robotics Expo (2023)